

In the first project assignment, you will be required to design the database together with the Data Definition Language (DDL) scripts to create that database. This project assignment will be performed in **pairs of students**. Each student has to contribute to the solution, otherwise, the other student can report the inactivity of his colleague, and then the colleague will be marked with 0 points from the assignment. The team members must consist of students from the same seminar group.

The assignment has no exact description, however, you need to meet the given requirements:

- Create a GitLab project repository named „bds-project-assignment-1“ inside your student space (https://gitlab.com/but-courses/bpc-bds/student-projects/2022/<YOUR_SPACE>/bds-project-assignment-1) and upload the database design together with all other files required below (each student will create their own repository). Include a suitable README.md file.
- The database must have at least 12 tables (same tables as are in seminar example project does not count; intersection tables counts).
- The database must have a one-to-one relationship.
- The database must have a one-to-many relationship.
- The database must have a many-to-many relationship.
- The database must be at least in the 3rd normal form.
- The database must contain an entity or entities for user sign-in purposes.
- For each table insert at least five records (for at least two tables insert more than 50 records).
- At least three tables must have more than five columns.
- The PostgreSQL database must contain some columns with the following data types: date, timestamp, serial/bigserial, decimal, boolean, varchar.
- Relationships between tables have to be carefully set as mandatory or optional.
- Database tables and columns should use consistent naming conventions (e.g., all table or column names as lowercase letters; when the name is multi-word, separate it by „_“ character).
- Database tables and columns must be named with English terms.
- **Optional task:** Integration of Flyway or Liquibase for the database versioning (create at least two versions of your database, e.g., the additional version will contain any ALTER TABLE command).

Together with these requirements, you have to provide a document with attachments that will contain the following:

- Picture representing the database design (e.g., Entity Relationship Diagram (ERD) with Crow's Foot Notation¹).
- Use-case diagram² representing system requirements with expected behavior.
- System context diagram³.

¹Crow's Foot Notation description: <https://support.microsoft.com/en-us/office/create-a-diagram-with-crow-s-foot-database-notation-1ec22af9-3bd3-4354-b2b5-ed5752af6769>

²Use-case diagram description: <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-use-case-diagram/>

³System context diagram description: <https://online.visual-paradigm.com/knowledge/system-context-diagram/what-is-system-context-diagram/>

- List of suitable non-functional and functional requirements⁴. Provide at least three functional and three non-functional requirements of your system.
- Application description (What type of application does your database represent? For example, a database for incident handling). Describe the use-case that you are trying to deal with.
 - At least 1/2 of A4 paper.
- Description of each table – the purpose, and one short descriptive sentence for each table column plus explanation why you choose a particular data type.
- Description why do you think that your database is in 3rd normal form?
- The DDL scripts to create a database for PostgreSQL database (use `.sql` extension to these DDL scripts).
- The DDL scripts to create a database for a different relation database of your choice (here, you need to modify your previously created DDL scripts to meet the requirements of another database).
- The evidence that you created a database (with all the tables) in PostgreSQL (e.g., screenshots from the pgAdmin).
- **Optional task:** Instructions on how to run the Flyway/Liquibase commands and what updates it performed.

Please upload all the documents with attachments into your GitLab repository project. This might be as follows:

- 1x Word/L^AT_EX document with all the required diagrams, application description, and other required information.
- 1x DDL script for PostgreSQL (DB schema).
- 1x DML script for PostgreSQL (DB data).
- 1x DDL script for an alternative database (e.g., MariaDB/MySQL).
- 1x DML script for an alternative database (e.g., MariaDB/MySQL).
- 1x README.md file with suitable information from Word/L^AT_EX document.

This project assignment is for 6 points.

- **Target date: 31th of October (inclusive)**
- 3 for satisfying all the requirements (if any is not met, the number of points is set to 0)
- 0-3 based on the quality of a solution

⁴Description of functional and non-functional requirements: <https://stackoverflow.com/a/16476014>